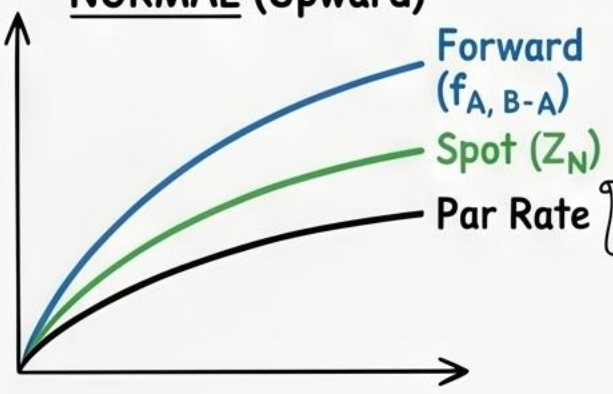
CFA Level II: The Term Structure & Interest Rate Dynamics (Module 1)

1. The Building Blocks—Three Essential Curves

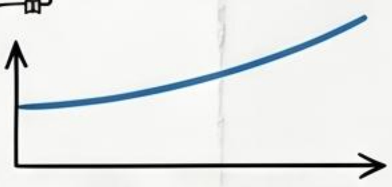
NORMAL (Upward)



Price of future payment

Coupon for Price=100

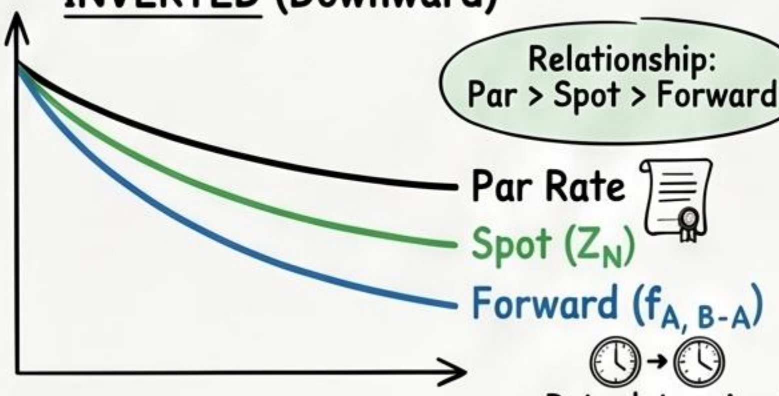
Relationship: Forward > Spot > Par



FLAT CURVE: Forward = Spot = Par

INVERTED (Downward)

Relationship: Par > Spot > Forward



Rate determined today for future

2. The Math Engine—Formulas & Bootstrapping

BOOTSTRAPPING

Solve Z_N iteratively using Par rates. Start with 1-year (Par=Spot), then plug into 2-year bond formula...

DISCOUNT FACTOR ($$DF_N$):

$$DF_N = \frac{1}{(1+Z_N)^N}$$

Price of $1 received in N periods

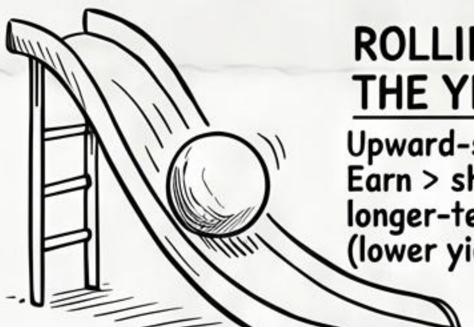
NO-ARBITRAGE VALUATION

$$\$Price = \left[\frac{CF_1}{(1+S_1)^1}\right] + \left[\frac{CF_2}{(1+S_2)^2}\right] + \ldots + \left[\frac{(CF_N + Principal)}{(1+S_N)^N}\right]$$

FORWARD RATE MODEL

$$(1+Z_B)^B = (1+Z_A)^A \cdot (1+f_{A,\,B-A})^{B-A}$$

3. Portfolio Strategies & Expected Returns



ROLLING DOWN THE YIELD CURVE

Upward-sloping, stable curve. Earn > short-term rate by holding longer-term bond as it ages (lower yields, higher prices)

EXPECTED vs. REALIZED RETURN

- If Spot = Forward prediction: All bonds earn 1-period risk-free rate
- If Spot remains unchanged (and upward sloping); Realized return > 1-period rate

SWAP SPREAD

Swap Spread = Risky Yield – Default-Free Yield

4. Why the Curve Bends—Term Structure Theories

PURE EXPECTATIONS: Forward rates = unbiased future spot predictors. Investors indifferent to maturity.

LIQUIDITY PREFERENCE: Investors demand 'term premium' for longer bands (interest rate risk). Explains upward slope.

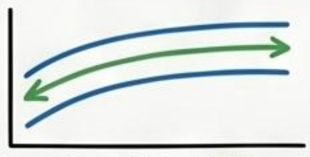
SEGMENTED MARKETS: Supply/demand in specific maturity buckets determine shape.

PREFERRED HABITAT: Similar to Segmented, but investors jump buckets for yield premium.

5. Dynamics & Risk Management

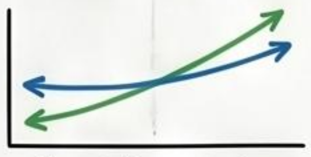
YIELD CURVE FACTOR MODEL (3 Movements)

1. LEVEL



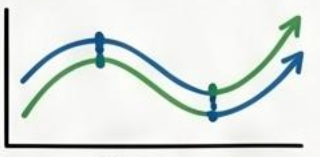
Parallel shift

2. STEEPNESS



Short/long rates move differently

3. CURVATURE



Intermediate rates move relative to ends

KEY RATE DURATION: Measures sensitivity to a shift in a specific curve point.

TED SPREAD: (3-m LIBOR/MRR – 3-m T-bill) High TED = Credit Risk & Low Liquidity

6. COMMON EXAM TRAPS (Pro-Tips)

- **YTM vs. SPOT RATES:** YTM is just an average IRR. In upward curve, coupon YTM < spot rate of final maturity.
- **THE FORWARD CURVE 'PULL':** Spot upward? Forward is ABOVE. Spot downward? Forward is BELOW.
- **BOOTSTRAP SEQUENCE:** Always solve for shortest maturity FIRST. Can't skip ahead!